Deep Learning-Based Medication Recommendation System for Primary Care Decision Support

Arnav Sahu
230960000

Hrithiq Gupta
230962154

Nakul Bhat
230960000

Pratyush Bhagat
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School of Computer Engineering
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The Problem with Medication Prescribing

Prescribing requires balancing:

- Therapeutic efficacy
- Contraindications & dosage
- Drug–drug interactions (DDIs)
- Patient comorbidities & demographics

EHRs create new opportunities for **data-driven clinical decision support**

The Problem with Medication Prescribing
Prescribing requires balancing:
Therapeutic efficacy
Contraindications & dosage
Drug–drug interactions (DDIs)
Patient comorbidities & demographics
EHRs create new opportunities for **data-driven clinical decision support**

Set the stage – prescribing is hard, mistakes happen, and EHRs give us the data to do something about it.

Surveyed Studies (1–7)

Study / Year	Arch.	Safety	Limitation
DP-CRNN (2021) [19]	CNN+RNN	No	Not drug-centric
Deep Patient (2016) [13]	Autoenc.	No	Static reps
Doctor AI (2016) [5]	RNN	No	Low interpretability
RETAIN (2016) [7]	Attn-RNN	No	Not drug-aware
GAMENet (2019) [15]	GNN+Mem	Yes	Dense EHR needed
SafeDrug (2021) [12]	Mol-GNN	Yes	Compute heavy
GRAM (2017) [6]	Ontology GAT	Partial	Ontology dependent

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Surveyed Studies (8–14)

Study / Year	Arch.	Safety	Limitation
LEAP (2017) [18]	Opt+Seq	Partial	Needs labels
MoleRec (2022–23) [2]	GNN	Yes	Complex training
ClinicalBERT (2019) [3]	Transformer	No	Pretrain cost
MemoryNet (2019) [16]	Memory NN	No	Needs history
HAN (2016) [17]	Hier. Attn	No	Complex
GRU-D (2018) [4]	GRU-Decay	No	Vitals focused
Polypharmacy GNN (2018) [20]	GNN	Yes	Graph sensitive

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Surveyed Studies (15–20)

Study / Year	Arch.	Safety	Limitation
Transfer Clinical NLP (2020) [11]	Pretrained	No	Domain shift
Multi-task CDSS (2019) [8]	Multi-DNN	Partial	ICU focused
RL Treatments (2018) [10]	RL	Partial	Reward design
LIME / SHAP [14]	Explainers	No	Approximate
KG Embeddings [1]	KG models	Yes	KG curation
Clinical Transformers (2022) [9]	Transformer	Partial	Resource heavy

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The Gap: Outpatient & Data-Sparse Settings

Existing models assume **rich longitudinal EHR data** (inpatient/tertiary care)

Outpatient reality:

- Sparse records, limited visit history
- Brief free-text symptom descriptions
- Often just **a single visit**

Safety is *even more* critical without longitudinal context

The Gap: Outpatient & Data-Sparse Settings
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Brief free-text symptom descriptions
Often just **a single visit**
Safety is *even more* critical without longitudinal context

This is the key motivation slide. Emphasize that sequence-dependent models just don't work here – and safety can't be an afterthought.

Our Approach

A hybrid deep learning architecture for **outpatient medication recommendation**

Key components:

- Attention-based symptom encoding
- Structured feature fusion
- DDI-aware safety filtering

Goals: **performance**, **safety**, and **interpretability**

Our Approach
A hybrid deep learning architecture for **outpatient medication recommendation**
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Attention-based symptom encoding
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Transition slide into the rest of the presentation. You can forward-reference the architecture diagram here if it comes up in the next section.

System Overview

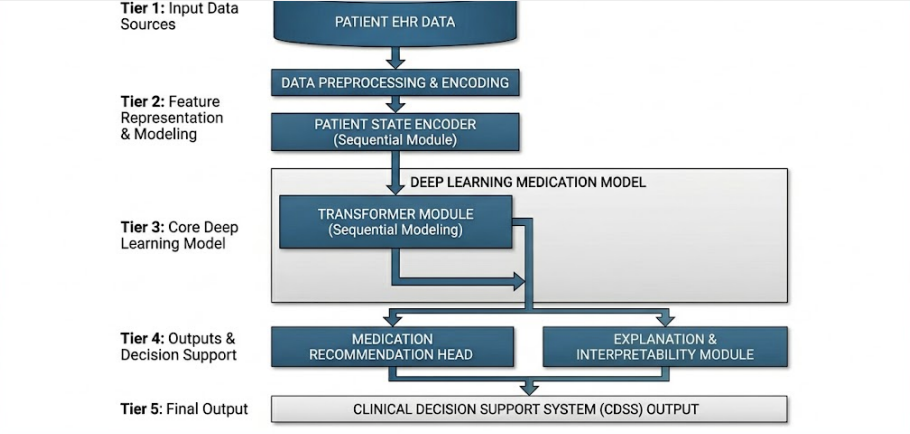
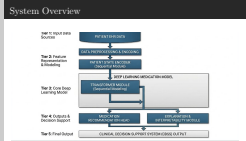


Figure: Four-phase medication recommendation pipeline.



Walk through the figure – four phases: preprocessing, representation learning, recommendation + safety, interpretability.

Preprocessing & Patient Representation

Input: Free-text symptoms + age, sex, comorbidities

- Spell correction, abbreviation expansion (e.g., SOB → Shortness of Breath)
- NER → SNOMED-CT / ICD-10 entity mapping
- **Symptom embedding E_s:** Clinical DistilBERT
- **Structured embedding E_d:** MLP over demographics & comorbidities
- **Patient state:** $P_v = \text{Concatenate}(E_s, E_d)$

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Stress that DistilBERT is lightweight by design – important for outpatient/low-latency settings.

Recommendation & Safety Filtering

Multi-label sigmoid head over $P_v \rightarrow$ per-drug-class probabilities

Label-correlation regularizer for co-prescription dependencies

Top-k candidates passed to **DDI-aware safety filter** (DrugBank graph)

Unsafe combinations $\rightarrow$ penalized in ranking or clinician alert

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Unsafe combinations $\rightarrow$ penalized in ranking or clinician alert

The safety filter is the key differentiator from vanilla multi-label classifiers. Mention the override alert for clinical trust.

Interpretability

SHAP saliency: token-level attribution over symptom text

Safety rationales: knowledge-graph trace for flagged DDIs
e.g., *"Drug A and Drug B both inhibit CYP3A4"*

Interpretability
SHAP saliency: token-level attribution over symptom text
Safety rationales: knowledge-graph trace for flagged DDIs
e.g., "Drug A and Drug B both inhibit CYP3A4"

Two layers of explainability – one for the recommendation, one for the safety alert. Both aimed at clinician trust.

Algorithm

Algorithm 1 DL Medication Recommendation

Require: S (symptom text), D (structured data), G (DDI graph)
Ensure: M_{safe}, explanation report E

1: Normalize S: spell correction + NER $\rightarrow$ ontology mapping
2: $E_s \leftarrow \text{ClinicalDistilBERT}(S)$; $\quad E_d \leftarrow \text{MLP}(D)$
3: $P_v \leftarrow \text{Concatenate}(E_s, E_d)$
4: Sigmoid classifier $\rightarrow$ top-k candidates
5: DDI safety filter over $G \rightarrow M_{safe}$
6: SHAP attribution $\rightarrow E$
7: **return** M_{safe}, E

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Condensed from the full algorithm – hit the key steps. Good slide to have open during Q&A.

Evaluation Plan

Data: Synthea synthetic outpatient records + DrugBank / TWOSIDES

Metrics:

- Recommendation: Jaccard index, mean average precision
- Safety: DDI occurrence rate
- Interpretability: clinician-rated explanation utility

Ablations: encoder variants, safety-layer removal, explanation fidelity

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We haven't run experiments yet – frame this as the planned evaluation. Be ready for questions on dataset size and clinician study design.

Dataset

Synthetic outpatient encounters built from multiple public sources: symptom–disease, drug–disease, DDI mappings + Synthea demographics

50,000 samples with multi-label medication outputs

Structured attributes: age (1–90), sex, comorbidity flags (diabetes, hypertension, asthma)

Text augmentation: synonym replacement, template sentences

DDI filtering + clinical heuristics (e.g., no ibuprofen for hypertensive patients)

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Structured attributes: age (1–90), sex, comorbidity flags (diabetes, hypertension, asthma)
Text augmentation: synonym replacement, template sentences
DDI filtering + clinical heuristics (e.g., no ibuprofen for hypertensive patients)

Stress that the dataset is synthetic but clinically grounded. The DDI filtering means ground-truth labels are already safe – important context for interpreting the DDI rate metric.

Models Compared

Model	Key Design Choice
MLP	Flattened token embeddings + structured features
TextCNN	1D filters over embeddings, local n-gram features
BiLSTM	Bidirectional sequence encoding
SelfAttention	Single-head attention + mean pooling
FusionNet	**LSTM (text) + MLP (structured), dual-branch**

Shared: 64-dim embeddings, max sequence length 20, BCE loss
Adam, lr $= 10^{-3}$, 5 epochs, batch size 32, 80/20 split

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The table cleanly shows the progression in complexity. FusionNet is the only model with an explicit dual-branch design – that's the hypothesis being tested.

Evaluation Metrics

Multi-label quality: Jaccard index, micro-F1, subset accuracy, Hamming loss

Ranking ($k = 5$): Precision@K, Recall@K, F1@K, PR-AUC

Discrimination: ROC-AUC (micro)

Safety: DDI rate – fraction of top-k pairs with known adverse interactions

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Discrimination: ROC-AUC (micro)
Safety: DDI rate – fraction of top-k pairs with known adverse interactions

Highlight the DDI rate – it's the only metric that directly measures safety, and it's what distinguishes this work from standard multi-label classification.

Training Convergence

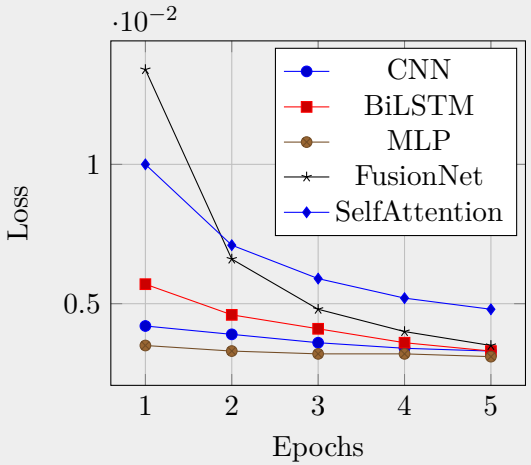
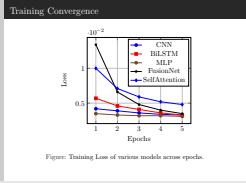


Figure: Training Loss of various models across epochs.



MLP lowest absolute loss but simpler model. FusionNet starts high, converges fast once text encoder stabilizes – key sign the dual-branch design works.

Model Performance — Micro and Macro Metrics

Model	Jaccard	Precision$_\mu$	Recall$_\mu$	F1$_\mu$
MLP	0.6887	0.8853	0.7561	0.8156
CNN	0.7657	0.9031	0.8342	0.8673
BiLSTM	0.7796	0.9218	0.8348	0.8762
SelfAttention	0.7684	0.9022	0.8382	0.8691
FusionNet	0.7869	0.9252	0.8404	0.8808
Model	**Hamming**	**Recall$_M$**	**F1$_M$**	**Precision$_M$**
MLP	0.0015	0.0675	0.0721	0.0781
CNN	0.0011	0.0762	0.0798	0.0858
BiLSTM	0.0010	0.0777	0.0821	0.0899
SelfAttention	0.0011	0.0753	0.0789	0.0864
FusionNet	0.0010	0.0786	0.0828	0.0898

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Model Performance — Error and AUC

Model	Subset Acc	ROC AUC	PRAUC
MLP	0.4210	0.9937	0.8372
CNN	0.5603	0.9968	0.8743
BiLSTM	0.6184	0.9957	0.8743
SelfAttention	0.5675	0.9970	0.8764
FusionNet	0.6455	0.9964	0.8753
Model	**Precision@K**	**Recall@K**	**F1@K**
MLP	0.6855	0.6329	0.5825
CNN	0.7101	0.6563	0.6042
BiLSTM	0.7076	0.6555	0.6028
SelfAttention	0.7114	0.6573	0.6052
FusionNet	0.7079	0.6561	0.6034

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Key Takeaways

FusionNet best overall – Jaccard 0.787, subset accuracy 0.646

SelfAttention best for top-k retrieval – highest PR-AUC & Precision@K

All models ROC-AUC > 0.993 – strong per-label discrimination

Beyond a representational threshold, **safety & interpretability** become the key differentiators

Key Takeaways
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All models ROC-AUC > 0.993 – strong per-label discrimination
Beyond a representational threshold, **safety & interpretability** become the key differentiators

Wrap up results cleanly. The last point is a nice segue into the safety filtering and SHAP discussion.

What the Results Tell Us

Complexity helps – but plateaus: FusionNet's **dual-branch fusion** is the key, not just model size

High ROC-AUC across all models: the symptom–medication signal is learnable even by shallow models

Subset accuracy gap (MLP 0.421 vs. FusionNet 0.646): recovering the *complete* drug set is hard – clinically, a single missed or unsafe drug matters

Low macro-F1 across the board: **rare drug classes** are systematically under-predicted

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Low macro-F1 across the board: **rare drug classes** are systematically under-predicted

The subset accuracy point is the most clinically meaningful – drive this home. Macro-F1 is a known limitation of the synthetic data, not a flaw in the model.

Limitations

DDI filter applied at **data construction**, not inference – no runtime safety measurement yet

Synthetic data: template-constructed text $\neq$ real clinical notes

Long-tail drug distribution: rare but clinically important medications under-predicted

Limitations
DDI filter applied at data construction, not inference – no runtime safety measurement yet
Synthetic data: template-constructed text ≠ real clinical notes
Long-tail drug distribution: rare but clinically important medications under-predicted

Be upfront here – the audience will ask. The DDI rate limitation is the most important one to acknowledge.

Conclusion

FusionNet best overall – modality-aware fusion suits outpatient multi-modal data

SelfAttention best for top-k shortlists – useful for clinician-facing interfaces

DDI-aware safety filter + SHAP interpretability address trust & safety beyond raw accuracy

Foundation for a **lightweight, safe, interpretable** primary-care decision support system

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Foundation for a **lightweight, safe, interpretable** primary-care decision support system

Keep this tight. The three adjectives – lightweight, safe, interpretable – are the pitch.

Future Work

- Evaluate on real-world clinical corpora
- Runtime DDI filtering with quantifiable rate reduction
- Class-imbalance strategies for rare drug prediction
- Larger clinical LMs (e.g., ClinicalBERT) as drop-in encoders in FusionNet

Future Work
Evaluate on real-world clinical corpora
Runtime DDI filtering with quantifiable rate reduction
Class-imbalance strategies for rare drug prediction
Larger clinical LMs (e.g., ClinicalBERT) as drop-in encoders in FusionNet

Good slide to end on – shows the work is a starting point, not a dead end. The LM swap is a quick win worth highlighting.

Authors' Contributions

Table: Dataset Generation Contribution

Task	Author
Disease–Drug Mapping	Arnav Sahu
Symptom–Disease Mapping	Pratyush Bhagat
Drug–Drug Interaction Mapping	Nakul Bhat
Dataset Combination & Generation	Hrithiq Gupta

Table: Model Contribution

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